

Adapting Before Contact:

Compositional Contextual Inference for Safe Control with Frozen World Models

Research Proposal — v4

(v3 merged with Direction 1: test-time data selection for frozen controllers)

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Abstract

Robots that plan with learned world models make mistakes when the physics changes, for example when the table surface becomes more slippery or the object becomes heavier. Current methods fix this by updating the model from its prediction errors. They can only do so after the robot has touched the object, and they discard what they learned when the next episode starts. For tasks that are decided at the moment of first contact, such as striking a puck toward a target or lifting an unknown load, these methods therefore give no benefit at all.

We propose a method based on a model of how humans learn motor skills [44]. The robot keeps a growing set of small correction models, one for each physical condition it has encountered (“memories”). It continuously estimates which condition it is currently in, and it learns without labels which visual features predict each condition. When a known condition returns, the robot selects the right memory from what it sees, before it touches anything. The memories share a prior that is organised by physical factor, such as surface and load. An unseen combination of known factors therefore gets a prediction before its first contact, together with a stated estimate of how wrong that prediction can be.

The method, Cira (Context-Indexed Residual Adapters), adds these memories to any frozen world model. The same estimate of the current condition drives a safety filter whose margins are tuned separately for each memory, and it decides whether a short probe before a strike is worth its cost. The proposal contains four theoretical results and one measurement, each tested on simulated and real robot tasks:

- the error at first contact grows with how unpredictable the sequence of conditions is, while for existing methods it grows with the number of episodes in which the condition differs from training;
- the memories keep valid confidence bounds while the condition is uncertain, with an extra error term controlled by the same unpredictability;
- a long-run bound on safety violations that holds whether or not the visual cues are reliable;
- a probe before contact has value only when the plausible conditions call for different actions, and the probing rule uses this;
- a method that splits the robot’s recovery after a change into “recalling an old memory” and “learning something new”, which we measure over a long real-robot deployment.

Changes from v3

This version merges v3 with Direction 1 (test-time data selection for adapting frozen controllers). The changes are:

1. **Memories share a prior across contexts** (Section 4, Eq. (2)). This is Direction 1’s multi-environment kernel applied to discrete context labels. It sets the starting point of every new memory, gives unseen combinations of known factors a memory slot before they are seen, and gives a predicted residual error for them (Proposition 4).
2. **Memories are updated with context probabilities computed before the current error is observed.** Theorem 5 gives confidence bounds for the memories under context

uncertainty, and it supplies the accuracy ε that Proposition 1 assumed. The v3 update, which used probabilities computed after the current error, biases the memories (Remark 1).

3. **Noise model.** v3 combined a matrix-normal prior with a state-dependent full noise covariance, which is not conjugate. v4 keeps independent rows with a diagonal state-dependent noise model. The principal directions decorrelate the training errors, which motivates the diagonal choice.
4. **Safety margins.** Each memory is now scored against its own prediction. v3 compared the smallest distance over all plausible memories with a per-memory margin, and with that rule the proof of Proposition 3 does not go through.
5. **Probing is chosen by value of information for the decision** (Proposition 8, H9). This is Direction 1’s decision-aware selection rule, applied to the probe. It is promoted from an optional extension to a contribution.
6. **From prediction error to task cost.** Direction 1’s simulation lemma (Lemma 6) turns first-contact prediction error into a bound on the loss in task cost (Corollary 7).
7. **Status labels** on every result.

Two parts of Direction 1 are not carried over. Selecting a subset of a data pool under a budget is not needed: conditioning a memory on all its data costs rank-one updates, and under a correct model more data never increases posterior variance, so a subset budget does not bind. The optimistic moving target pays off only when data gathered along the plan are used before the decision is final. At first contact they cannot be (Section 5). The convergence of transductive selection under a moving target remains an open question and is kept as separate work.

Status labels. [Proved] an established result, cited. [Proved here] an argument given in full in this document. [Provable with effort] a route is clear but the details are not written. [Heuristic] a motivated approximation without a guarantee.

1 Motivation and contributions

World models and their adaptation. A latent world model encodes camera images into a compact vector (the latent state) and predicts how that vector changes when the robot acts. Planners use these predictions to choose actions. Such models now plan manipulation tasks without task-specific training [8, 39, 93, 110]. When the physics at deployment differs from the training data, the predictions become wrong. Two recent methods address this. AdaJEPA updates part of the model with a few gradient steps on its recent prediction errors while the robot runs [102]. Sandwich-Residuals keeps the model fixed and trains small correction modules around it [94]. Both reset to the original model at the start of every episode, and both list keeping what was learned across episodes as future work. Keeping the gradient updates running indefinitely is not a good alternative. On long streams of data such updates gradually drift and degrade, and simply resetting from time to time works better [48, 85].

Real deployments repeat themselves. A robot in a lab, warehouse or kitchen rarely meets a completely new condition. It meets the same few surfaces, containers and tool states repeatedly, and these usually look different from each other. Humans make use of this. Research on human motor learning shows that people keep separate motor memories for different conditions. They recall a memory when a condition returns and do not relearn it (savings). Learning a new condition does not erase the old one. Visual cues can select the right memory before the movement starts [43, 44, 45].

Running example. A robot arm strikes a puck toward a target on a table. The table is covered with one of three mats: felt (high friction), rubber (medium) or acrylic (low). The puck is sometimes

empty and sometimes weighted. The mats and weights are swapped by people during the day. If the robot assumes the wrong friction, the puck stops short or slides off the table edge. The strike is a single contact, so the robot has no chance to correct its action after seeing the error.

Main idea. We treat adaptation as the question “which of the conditions I know is active now, or is this a new one?”. The robot answers this question with a probability distribution over its memories (the context posterior). It updates mainly the memory that is likely to be active. This matters most at the first contact. At that point error-driven methods have received no error signal. The outcome of single-contact tasks is decided there, and so is much of the risk of unsafe behaviour.

Contributions.

- C1. **Single-contact tasks as a test.** We define a class of tasks in which methods that only learn from prediction errors cannot outperform the unadapted model (Observation 1). Examples are striking an object toward a target, lifting an unknown load and pushing near a table edge. On these tasks, only information available before contact can help.
- C2. **The method, Cira.** A set of small probabilistic correction models (memories) on top of a frozen world model, with closed-form updates. One probability distribution over memories decides when a new memory is created, which memory is used and updated, and when the robot has returned to normal conditions. The memories share a prior organised by factor (for example surface and load), so an unseen combination of known factors is predicted before contact. A noise model that depends on the state prevents the short, noisy moments at contact from being mistaken for a change of condition.
- C3. **Theory.** Proposition 1 bounds the total error at first contact. For our method the bound depends on how predictable the sequence of conditions is given the visual cues. For reset-based methods it grows with every episode that differs from training. Theorem 5 shows that memories updated under context uncertainty keep valid confidence bounds, up to a term controlled by the same kind of log-loss, and so supplies the memory accuracy that Proposition 1 assumes. Proposition 4 states when an unseen combination is predicted exactly and gives the residual error otherwise. Corollary 7 turns first-contact prediction error into a bound on the loss in task cost. Proposition 2 predicts how many steps the robot needs to identify a known condition. We state these predictions before running the experiments.
- C4. **A safety filter that uses the memories.** The filter checks each planned action against every memory that is still reasonably likely. Its safety margin is tuned online, separately for each memory. When a known condition returns, the robot therefore also recovers the margin that was appropriate for it. Proposition 3 bounds the long-run rate of safety violations for any sequence of conditions, including when visual cues are misleading.
- C5. **Probing when it pays.** Before a strike the robot can give the object a small nudge. It does so only when the nudge is expected to change the chosen strike by enough to pay for itself. Proposition 8 shows that a probe has zero value whenever all plausible conditions call for the same strike, however uncertain the robot is about which condition is active. A probing rule based on information about the condition probes in exactly those cases.
- C6. **A measurement and a long deployment.** We split the improvement after a change of condition into two parts (Definition 2): the part that comes from selecting an existing memory, and the part that comes from changing a memory’s parameters. Following the human motor learning literature, we call these apparent and proper learning. We report this split over a real-robot deployment of at least 1000 episodes, in which people swap surfaces and containers on a schedule the robot does not know. We also release a benchmark of long sequences of changing conditions, built around four classic motor-learning experiments.

Scope. We study changes in physics (friction, mass, load, motor strength) while the image encoder stays fixed. Appearance matters only as a clue about which condition is active. We do not try to correct for changes in appearance that confuse the encoder itself, which is what AdaJEPA and Sandwich-Residuals address. We also assume that a safety constraint expressed in the latent space matches the real constraint. Whether it does is a property of the encoder that we do not study here [72].

2 Related work

Contextual inference in human motor learning. The MOSAIC model proposed that the brain holds several paired prediction and control models and chooses between them using both prediction accuracy and signals available before a movement [43, 103]. The COIN model [44] turned this into a full probabilistic model. It has a set of contexts that can grow without limit and that tend to persist over time, one simple memory per context, and visual or other cues whose probability depends on the context. COIN explains savings, interference between opposing tasks, and the spontaneous return of old behaviour through changes in which memory is used, with learning speed unchanged. It also predicted two effects that the standard two-rate model [92] cannot produce, and experiments confirmed both: the return of an old memory triggered by a few reminder trials (evoked recovery), and a cue-dependent amount learned per trial. Related models of animal learning make the same distinction between updating a memory and creating a new one [34, 35, 84]. People also learn separate parts of a task independently and recombine them [31].

One result is a warning for our design. Many purely visual cues do not help people separate motor memories [49], while differences in the planned movement do [90]. We therefore do not assume that a cue is useful. The robot learns how informative each cue is, and we vary cue reliability in the experiments.

Adapting world models while they run. AdaJEPA [102] updates encoder and predictor layers with about one gradient step per planning cycle on the PushT, PushObj and PointMaze tasks, including changes in mass and damping. Its authors note that adaptation is limited by what the pretrained representation can express. Sandwich-Residuals [94] reaches similar results while training only 1–3% as many parameters. More general test-time adaptation methods [96, 99, 101] degrade on long and repeating data streams [48, 85]. Contextual world models estimate a continuous context vector from recent experience [11, 62, 86]. Change-point-aware world models discard outdated training data after a detected change [106]. None of these methods stores separate conditions and recalls them later, and none acts before contact.

Probabilistic tools for switching dynamics. Our method combines existing tools: Bayesian last layers, which give fast closed-form updates of the final layer of a network [40, 42]; methods that detect when the data-generating process changes [2, 41]; mixtures of dynamics models whose number grows with the data, using a Chinese restaurant process (CRP) prior [53, 63, 79, 105]; switching linear dynamical systems [29, 30, 67]; the interacting multiple-model (IMM) filter from target tracking [15, 65, 76]; and lifelong learning with factored or modular models [28, 56, 77]. The closest prior method is MOLe [79], which keeps a CRP mixture of dynamics models for model-based reinforcement learning. It selects a model only from prediction errors, so it cannot act before contact. In online learning theory, “mixing past posteriors” [18] handles a small set of recurring experts. Our first-contact result builds on the same intuition.

Transfer between environments and decision-aware data. Multi-output kernels share statistical strength between related tasks; the separable (intrinsic coregionalisation) form puts all transfer into a kernel over tasks [6]. Transductive active learning selects data by their information about a target set that need not overlap the data [51], and active fine-tuning of multi-task policies applies this to choosing which task demonstrations to collect [9]. We use the same kind of task kernel

	Latent visual world model	Keeps / recalls memories	Acts before contact	Cue learned during use	Combines factors	Safety uses memories	Calibrated under change
AdaJEPA [102], Sandwich-Res. [94]	✓	–	–	–	–	–	–
MOLe [79], IMGP [105]	–	✓	–	–	–	–	–
MOCA [41]	–	keeps only	–	–	–	–	partly
VcVBLL [19], Phys2Real [100]	–	–	✓	– (fixed in advance)	–	–	✓
Uncertainty-aware latent filter [89]	✓	–	–	–	–	avoids unknown	✓
Latent world model + ACI filters [52, 82]	✓	–	–	–	–	one model	✓
COIN [44] (model of humans)	–	✓	✓	✓	–	–	–
Cira (this proposal)	✓	✓	✓	✓	✓	✓	✓

Table 1: Comparison with the closest work, based on literature searches in September 2026. Each column is covered by some prior work. Our contribution is the combination, the single-contact task class and the first-contact analysis.

over discovered contexts, where it sets the prior of every new memory. For information gathering we take the decision itself as the target. In the linear, parametric setting this is identification for control and least-costly experiment design [17, 36, 47], whose greedy form is Elfving’s c -optimal design [27]. In the discrete setting it is value-of-information analysis [50, 57]. Confidence bounds for regression on adaptively collected data rest on self-normalised martingale inequalities [1, 22]. Weighted [88] and misspecified [61] versions exist; Theorem 5 combines the two for weights set by a context filter.

Using vision to predict dynamics in advance. VcVBLL [19] adjusts a vehicle dynamics model using a hand-designed score for standing water in the camera image. Terrain-aware models use pretrained image features in a similar way [3, 20, 38]. Other methods train an image model online to predict physical properties measured by the robot’s own sensors [32, 69, 75]. Vision-language models can also provide prior estimates of physical properties [33, 100]. All of these map an image to a continuous physical quantity through a channel chosen in advance. Cira maps the image to one of the robot’s own discovered conditions, and lets the observed dynamics overrule the image when they disagree.

Safety in latent space. Several methods compute safety filters directly in the latent space of a world model, using reachability analysis or control barrier functions (CBFs) [7, 72, 80, 81, 104]. A CBF is a function h that is positive in safe states. A controller stays safe if it keeps h from decreasing too quickly. Seo et al. [89] treat states outside the training data as unsafe and steer away from them. AnySafe changes the safety constraint at runtime but not the dynamics [5]. Conformal prediction gives error bounds for learned predictors, and adaptive conformal inference (ACI) keeps these bounds valid when the data change over time [25, 37, 52, 66, 82, 95]. Adaptive and robust CBFs narrow down a single set of unknown parameters online [21, 68, 97, 98]. Multi-stage model predictive control is robust against a fixed set of candidate models [71]. What is missing is a filter whose set of candidate models is the set of remembered conditions that are currently likely. Such a filter can be cautious in new conditions, less cautious in recognised ones, and can store its calibration with each memory.

3 Problem setting

A frozen encoder maps an image o_t to a latent state $z_t = E(o_t) \in \mathbb{R}^d$. A frozen predictor gives the next latent state $\hat{z}_{t+1} = f(z_t, a_t)$ for action a_t . The robot runs one long sequence of episodes. It knows when an episode starts and ends, but it is never told which condition is active, and its memories are never reset.

In episode e an unknown context c_e^* is active; it occasionally changes within an episode, and we write c_t^* for the context at step t . A context is a physical condition, such as “acrylic mat, weighted puck”. Under context c , the true next state differs from the frozen prediction by an average amount $\mu_c(z, a)$, which we call the residual. The training condition has zero residual, $\mu_0 \equiv 0$. Before the first contact, at time t_c , the robot sees the scene and extracts a visual cue y_e . Before contact, the robot’s own motion reveals nothing about the context. We write $\mathcal{Z}_A$ for the set of latent state–action pairs that occur at and after contact under the candidate plans, and define

$$\Delta_{\max} = \max_{c, c'} \sup_{(z, a) \in \mathcal{Z}_A} \|\mu_c(z, a) - \mu_{c'}(z, a)\|, \quad \Delta_{\min} = \min_{c \neq 0} \inf_{(z, a) \in \mathcal{Z}_A} \|\mu_c(z, a)\|.$$

Definition 1 (Single-contact task). A task is single-contact if its success, or a violation of its safety constraint, is decided by the prediction the model makes at the first contact t_c and the action chosen from it.

Examples are striking a puck toward a target, tossing an object [109], other fast non-grasping motions [73], lifting an unknown load without jerking or slipping, and pushing an object near a table edge on an unknown surface.

Observation 1 (Error-driven adaptation cannot help at first contact). Suppose a method changes its model only in response to prediction errors. Since there are no errors before contact, its prediction at t_c is the same as at the start of the episode. Methods that reset every episode (AdaJEPA, Sandwich-Residuals, gradient updates with reset) therefore make the frozen model’s prediction at first contact. Their error there is $\|\mu_{c_e^*}\|$. On single-contact tasks they cannot do better than the unadapted model.

The observation follows directly from the definitions. Its purpose is to make clear that on these tasks only two sources of information remain: what the robot has seen in past episodes, and what it sees in the scene before contact.

4 Method: Context-Indexed Residual Adapters

Overview. Cira has five parts: a small set of directions in the latent space where corrections are made; one correction model (memory) per known context; a prior that links the memories through the physical factors they share; a probability distribution over which memory is active, updated at every step; and a learned link between visual cues and memories. The world model itself is never changed.

Where corrections are made. We collect the prediction errors $r_t = z_{t+1} - f(z_t, a_t)$ of the frozen model on training-condition data. We keep the k directions along which these errors vary most (principal components, with k between 4 and 16), stored as a matrix $U \in \mathbb{R}^{d \times k}$ with orthonormal columns. All corrections are made in these k directions, using the projected error $e_t = U^\top r_t$. The reasoning is that a few physical parameters, such as friction and mass, change the dynamics along only a few directions [26, 56].

Memories. Each context c has a memory: a linear model that predicts the projected error from a small feature vector $\phi_t = \phi(U^\top z_t, a_t) \in \mathbb{R}^p$, for example $\phi_t = [a_t, U^\top z_t, 1]$. For each output coordinate $i = 1, \dots, k$,

$$e_{t,i} \mid c \sim \mathcal{N}(w_{c,i}^\top \phi_t, \sigma_i^2(z_t, a_t)), \quad (1)$$

and each row $w_{c,i}$ has a Gaussian posterior, so the memory can be updated in closed form after every step [40]. The principal directions are uncorrelated on training data, which motivates independent rows and a diagonal noise model. The model includes the action, because a change in friction or motor strength produces an error that grows with the action. A constant offset could not describe this. The correction is also linear in the action, which the safety filter in Section 5 uses. Optionally the weights may drift slowly, so a memory keeps learning while it is in use; the theory below treats fixed weights (drift is discussed after Theorem 5). The memory for the training condition is fixed at $W_0 = 0$.

Noise that depends on the state. The noise scales $\sigma_i(z, a)$ come from a small network trained on training-condition errors [55]. Around contact the frozen model is already less accurate, for example at the moment of impact or during stick-slip motion, and σ_i is larger there. Large errors at those moments are then treated as expected noise, and they do not count as evidence of a new condition. Pushing is known to have this kind of state-dependent noise and variable friction [10, 74]. We found no prior work that links contact noise to the false creation of new contexts. We expect it to be the most common failure of a simple implementation, and our simulated tasks include it on purpose.

Memories share a prior. A context consists of several factors, such as the surface and the load: $c = (c^1, \dots, c^F)$. Contexts that share a factor value should share part of their correction. We place a Gaussian prior across memories, row by row:

$$\text{Cov}(w_{c,i}, w_{c',i}) = k_C(c, c') \Sigma_0, \quad k_C(c, c') = \sigma_0^2 + \sum_{f=1}^F \sigma_f^2 \mathbb{I}[c^f = c'^f] + \sigma_\times^2 \mathbb{I}[c = c']. \quad (2)$$

The first term is a shift common to all conditions, the middle terms give one component per factor value, and $\sigma_\times^2$ is the part specific to each combination, which absorbs interactions such as friction growing with the normal force. This is the intrinsic coregionalisation model [6] with a linear input kernel $\phi^\top \Sigma_0 \phi'$, applied to discrete context labels. The training condition enters as a noise-free observation $W_0 = 0$ at its own factor values. The hyperparameters are set by maximising the marginal likelihood of the stored memories under weakly informative priors; with few contexts this estimate is uncertain, and we report its spread.

Each memory keeps its own posterior. Its prior centre m_c and prior covariance are recomputed at the start of each episode as the conditional under (2) given the current posteriors of the other memories, each treated as a Gaussian observation of that memory’s weights. [Heuristic] This is an approximation to joint inference over all memories; it is exact when the other memories are known exactly. The shared prior has three uses.

- A new memory starts from the conditional given the stored ones. The part of the new context that the stored contexts Ω cannot explain is the conditional variance $s_c = k_C(c, c) - k_C(c, \Omega) K_\Omega^{-1} k_C(\Omega, c)$. We report s_c for every new context. It separates what transfer from stored memories can achieve from what on-context data must supply.
- An unseen combination of known factor values, such as the acrylic mat with the weighted puck, gets a memory slot before it is ever seen. Its prior centre is the prediction used before its first contact. Proposition 4 states when this prediction is exact.
- v3 kept factor memories and combination memories separately, and added factor memories to start an unseen combination. Under (2) these are one model: each seen combination has one memory, and the factor structure lives in the prior.

Two things let the robot separate the factors without labels. Each factor has its own image region: the surface cue comes from the table area around the object, and the load cue from a crop of the object or container. Factors also change at different times: people swap mats and containers independently, so the robot sees one factor change while the other stays the same.

Estimating the active context. At every step the robot updates a probability distribution π_t over the slots: the stored memories, the unseen combinations of known factor values, the training condition, and one “new value” slot $\star_f$ per factor, whose other factor values are taken from the most likely known values under the cue. Three sources of evidence are combined. Before an episode, a context that has been seen often is considered more likely (the Chinese restaurant process prior [83], which gives each known context a probability proportional to how often it has occurred and a new context a small fixed probability). The visual cue enters through the cue model described below. Within the episode, each memory’s prediction of the latest error enters through Eq. (1). Contexts are assumed to persist, with a small chance of switching [30]. The update follows the interacting multiple-model filter [15]:

$$\pi_{t|t-1}(c) = \sum_{c'} p(c | c') \pi_{t-1}(c'), \quad \pi_t(c) \propto p(e_t | c, \phi_t) \prod_f p(y^f | c^f) \pi_{t|t-1}(c), \quad (3)$$

where the cue factor is applied once per episode. This one distribution answers four questions, so no separate change detector is needed: if a new-value slot stays most likely for several steps, a new memory is created; if a stored memory is most likely, it is recalled; if the training condition is most likely, the robot has returned to normal; if the distribution shifts, the condition has changed. Two memories that become nearly identical are merged.

Updating the memories. Memory c is updated with the error e_t at weight $\pi_{t|t-1}(c)$, the probability of c before e_t is seen. In the Gaussian update of row i this means the likelihood variance $\sigma_i^2(z_t, a_t)$ is divided by $\pi_{t|t-1}(c)$. v3 used $\pi_t(c)$, the probability after e_t is seen. With that choice the current error decides where it is stored: errors that look unusual for the active memory are sent elsewhere, and the active memory learns from a truncated error distribution. Remark 1 gives the size of the resulting bias, and Theorem 5 shows that the weights $\pi_{t|t-1}$ keep the confidence bounds valid. The price is that the step on which a change occurs is stored with the old memory.

Learning what cues mean. For each factor value, the robot keeps a Gaussian model $p(y^f | c^f)$ of what the cue looks like under that context. After each episode, this model is updated using context probabilities computed from the dynamics only, without the cue. If the cue model were instead trained on decisions that the cue itself had influenced, a wrong cue could reinforce itself. With this rule, a misleading cue is gradually unlearned.

Optional: priors from vision-language models. A vision-language model that estimates physical properties [33, 100] could set the starting guess for a new context from the image. This would help even the first time a condition is seen. We treat this as an extension and keep it separate from the main claims.

5 Planning, probing and safety in latent space

Planning. We use a sampling-based planner, the cross-entropy method (CEM) or MPPI [23, 87]. Each sampled action sequence is simulated with the frozen model plus a correction, $\hat{z}_{t+1} = f(z_t, a_t) + U\hat{W}_c\phi_t$, for each context c with non-negligible $\pi_t(c)$. The planner minimises the average predicted cost over contexts, $\hat{J}(a) = \sum_c \pi_t(c) \hat{J}_c(a)$, or a risk-averse version (CVaR) that focuses on the worst outcomes [64, 107]. Updating π_t and the memories costs much less than one call of the world model.

Why the planner is not optimistic. In model-based reinforcement learning, planning under the most favourable plausible model controls the suboptimality term that lives on the unknown optimal policy’s trajectory, and it pays off because data collected along the optimistic plan improve the model before later decisions [24, 54]. At first contact nothing learned during the strike can change the strike. We therefore bound the decision gap uniformly over candidate strikes (Corollary 7), plan

with the expected cost or its CVaR, and gather information only through an explicit probe whose value is computed.

Decision-aware probing. Before a strike, the robot may spend a short probe p , such as a small nudge, whose outcome o updates the context distribution. For a distribution b over contexts let $V(b) = \min_{a \in \mathcal{A}} \sum_c b(c) \hat{J}_c(a)$, where $\mathcal{A}$ is the set of elite strikes from CEM. The value of information of p is

$$\text{VOI}(p) = V(\pi) - \mathbb{E}_{o \sim p(\cdot|p, \pi)} [V(\pi_o)], \quad (4)$$

where π_o is the context distribution after observing o . Among probes that satisfy the safety condition (5), the robot picks the one maximising $\text{VOI}(p) - \kappa(p)$, where κ is the cost of probing (time and task penalty), and it probes only if this maximum is positive. The costs $\hat{J}_c(a)$ are already computed by the planner for the elite strikes, so (4) needs one Monte Carlo loop over outcomes. [Heuristic] This reweights contexts and ignores within-memory learning from the probe, and it ignores the small change of state that the probe causes. When one context dominates, the remaining uncertainty is within its memory. For that case we use the linearised criterion

$$\Delta(p) = \frac{(g^\top \Sigma \phi_p)^2}{\phi_p^\top \Sigma \phi_p + \sigma^2(z_p, a_p)}, \quad g = \nabla_w [\hat{J}(a_1) - \hat{J}(a_2)],$$

written for one output row with posterior covariance Σ , where a_1, a_2 are the two best strikes. This is the posterior variance reduction of the cost difference that decides between them: greedy c -optimal design [27], the linearised form of least-costly experiment design [17, 47]. [Heuristic] It fails where the cost is strongly nonlinear in w , for example near the table edge. The baseline rule of v3 picks the probe with the largest $I(c; o | p)$. Proposition 8 shows where the two rules differ.

Which contexts the safety filter considers. The planner uses the full distribution π_t . The safety filter uses only the contexts with probability at least δ , the plausible set $P_\delta = \{c : \pi_t(c) \geq \delta\}$, with δ below one over the number of slots so that P_δ is never empty. This set includes a new-value slot whenever that slot’s probability exceeds δ . When the robot recognises a known condition, P_δ contains a single memory. When it is unsure, or the condition looks new, P_δ contains several, and the filter becomes more cautious. As a baseline we also consider a filter that guards against all stored memories at once, as in multi-stage model predictive control [71]. We expect that baseline to be much more cautious.

The safety condition. We assume a latent safety function $h(z)$ is given, for example a latent CBF or reachability value trained with existing methods [80, 81]. States with $h(z) \geq 0$ are safe. The filter accepts an action a only if the discrete-time CBF condition [4] holds for every plausible memory:

$$h(f(z_t, a) + U\hat{W}_c\phi(z_t, a)) - L_h q_t^c \geq (1 - \alpha) h(z_t) \quad \text{for all } c \in P_\delta. \quad (5)$$

In words, h must not drop by more than a fraction α in one step under any plausible memory, with an extra safety margin $L_h q_t^c$. Here L_h bounds how fast h can change, and q_t^c is an estimate of how wrong memory c can be, explained below. Because the correction is linear in the action, linearising f and h around a planned action turns the condition into one linear constraint per plausible memory. The filter is then a small quadratic program. Over the planning horizon we apply the same condition by discarding sampled action sequences that violate it. If no action satisfies it, the robot stops or retracts.

Safety margins stored with each memory. After each step we compute, for each plausible memory, how far the actual latent state was from that memory’s prediction, in the full latent space: $s_t^c = \|r_t - U\hat{W}_c\phi_t\|$. This distance includes ordinary noise, errors outside the k correction directions and mistakes in identifying the context. If some plausible memory covered the step ($s_t^c \leq q_t^c$), the step is attributed to the most likely such memory. Otherwise it is attributed to the most likely

plausible memory and counts as a miss. Only the attributed memory’s margin is updated, by adaptive conformal inference (ACI) [37]:

$$q^c \leftarrow q^c + \gamma(\mathbb{1}[s_t^c > q_t^c] - \delta_s).$$

ACI raises the margin after a miss and lowers it slightly otherwise, so that in the long run the margin is exceeded on a chosen fraction δ_s of the memory’s steps. When a known condition returns, its margin is therefore already tuned and does not have to be learned again.

What happens when the cue is wrong. The cue affects π_t and therefore P_δ . A reliable cue shrinks the plausible set before contact, so the robot can act less cautiously from the start. A misleading cue can leave the true context out of P_δ . In that case the measured distances grow and ACI raises the margin. The long-run bound in Proposition 3 still holds. A wrong cue therefore costs extra caution and a limited number of extra violations. We measure this cost against a filter that ignores cues.

6 Theory

6.1 Assumptions

Assumptions marked * do real work and can fail in practice.

- A1* (*Realisable memories.*) For every context c there is $W_c \in \mathbb{R}^{k \times p}$ with $\mu_c(z, a) = W_c \phi(z, a)$ on the inputs that occur, constant in time. Misspecification of the feature map or of the correction directions adds a bias to every bound below.
- A2* (*Noise.*) $e_t = \mu_{c_t^*}(z_t, a_t) + \varepsilon_t$, and given $\mathcal{F}_{t-1}$ each coordinate $\varepsilon_{t,i}$ is zero-mean and $\sigma_i(z_t, a_t)$ -sub-Gaussian. Here $\mathcal{F}_{t-1}$ contains all observations up to and including z_t, a_t and the episode cue, and also the current context c_t^* . The context sequence may therefore be adaptive or adversarial, but it may not depend on the current noise. The noise model σ_i must not underestimate the true noise; this is where a poorly fitted noise network breaks calibration.
- A3* (*Lipschitz; used only for task-cost results.*) The true latent map under any plan considered is L -Lipschitz, the stage cost is L_c -Lipschitz, and h is L_h -Lipschitz. Write $C_H = \sum_{j=0}^{H-1} L^j$. In the latent tasks these constants are unknown, so the task-cost results are quantitative only in the state-based simulations.
- A4 (*Analysis rollouts.*) For Lemma 6 and Corollary 7, rollouts are noise-free, $z_{t+1} = f(z_t, a_t) + U\mu_{c^*}(z_t, a_t)$. Removing this needs an expectation over noise; this is routine but notationally heavy.
- A5 (*Safe fallback; used only in Proposition 3.*) Whenever no action satisfies (5), the fallback action satisfies $h(z_{t+1}) \geq (1 - \alpha)h(z_t)$.

6.2 First contact and recognition

Proposition 1 (Error at first contact). *Consider N episodes whose contexts come from a finite set of K contexts, visual cues that take values in a finite set $\mathcal{Y}$, and memories whose mean predictions are accurate to within ε on $\mathcal{Z}_A$ (Theorem 5 supplies such an ε with high probability).*

- (i) [*Proved here*] Any method that resets every episode has total first-contact error $\sum_e \|\mu_{c_e^*}\| \geq N_{\text{shift}} \Delta_{\min}$, where N_{shift} is the number of episodes whose context differs from training.
- (ii) [*Proved here*] In episode e , Cira’s probability-weighted prediction has first-contact error at most $\varepsilon + (1 - \pi_0^e(c_e^*))\Delta_{\max}$, where $\pi_0^e(c_e^*)$ is the probability it assigns to the true context before contact. Summed over episodes, $\sum_e (1 - \pi_0^e(c_e^*)) \leq \sum_e -\log \pi_0^e(c_e^*)$.

(iii) [Proved, standard; the creation term is provable with effort] The right-hand side of (ii) is the total log-loss of predicting each episode’s context from its cue. For any sequence of contexts it is at most $N\hat{H}(C | Y) + O(K|\mathcal{Y}|\log N)$ [59], where $\hat{H}(C | Y)$ is the empirical conditional entropy of the context given the cue.

Cira therefore has lower total first-contact error than any reset-based method whenever $\varepsilon + \Delta_{\max}\hat{H}(C | Y) < \Delta_{\min}N_{\text{shift}}/N$ (entropy in nats). This ignores lower-order terms and the $O(K)$ episodes in which memories are first learned. It holds for any order of contexts, including adversarial ones.

Proof sketch. Part (i) follows from Observation 1. For (ii), write the prediction as $\sum_c \pi(c)\hat{\mu}_c$. Its error is at most $\sum_c \pi(c)\|\hat{\mu}_c - \mu_c\| + \|\sum_{c \neq c^*} \pi(c)(\mu_c - \mu_{c^*})\| \leq \varepsilon + (1 - \pi(c^*))\Delta_{\max}$, and $1 - x \leq -\log x$ for $x \in (0, 1]$. Part (iii) is the Krichevsky–Trofimov bound for sequential prediction with a Dirichlet mixture, applied separately to each cue value. Creating new contexts adds $O(K \log N)$. $\square$

Interpretation. The conditional entropy $\hat{H}(C | Y)$ measures how hard it is to guess the context of an episode from its cue and from past frequencies. If conditions repeat and the cue identifies them, it is close to zero and Cira makes almost no first-contact error. If every episode brings a new condition, memory cannot help and the bound says so. The benefit of the cue is the reduction $\hat{H}(C) - \hat{H}(C | Y)$, which is the information the cue carries about the context. The same quantity explains three effects: savings, correct action before contact, and the lack of benefit from an uninformative cue.

Proposition 2 (Steps needed to recognise a known context). [Proved under i.i.d. increments (Wald’s identity); heuristic otherwise] Consider an episode whose true context c^* is stored in memory, and let c' be the most likely alternative. Suppose that after contact each step adds, on average, $D > 0$ to the log-odds in favour of c^* , where D is the KL divergence between the two memories’ predictions, and that the step contributions are independent. Before contact they are zero. The starting log-odds combine past frequency and cue: $\lambda_0 = \log \frac{n_{c^*}}{n_{c'}} + \log \frac{p(y|c^*)}{p(y|c')}$, where n_c counts past occurrences of c . If λ_0 already exceeds the decision threshold λ^* , the number of steps needed is zero. Otherwise the expected number of steps is at least $(\lambda^* - \lambda_0)/D$, with equality apart from the final overshoot.

The increments are neither independent nor identically distributed in practice, because D depends on the state; the proposition is used for predictions, which the experiments check:

- *Acting before contact.* A learned cue that is correct with probability ρ lets the robot correct about a fraction ρ of the error before contact.
- *Savings.* Each further occurrence of a context shortens recognition by $\log(n_{k+1}/n_k)/D$ steps, so the gains shrink over time. For gradient updates with reset, the time stays the same.
- *Misleading cues.* A misleading cue of strength ℓ_y (its log-likelihood ratio) costs about $2\ell_y/D$ extra steps compared with a correct cue.
- *Noise.* The state-dependent noise model lowers D around contact. This makes recognition slower but prevents false new contexts. We measure both effects. For the case without cues, classical results on detection delay [60, 70] serve as a reference.

6.3 Long-run safety

Proposition 3 (Long-run safety for any cue quality). [Proved here, using the deterministic form of the ACI argument [37]] Suppose (5) is enforced whenever some action satisfies it, and A5 holds otherwise. Suppose the scores are bounded, $0 \leq s_t^c \leq \bar{S}$, the initial margins lie in $[0, \bar{S}]$, and each memory runs ACI with target δ_s and step size γ on the steps attributed to it, as in Section 5. Let K_T be the number of memory slots used up to step T . Then, for any sequence of contexts and cues, the one-step condition $h(z_{t+1}) \geq (1 - \alpha)h(z_t)$ fails on at most

$$\delta_s T + K_T \frac{\bar{S} + 2\gamma}{\gamma}$$

of the first T steps.

Proof. Covered steps are safe. Suppose step t is attributed to $c \in P_\delta$ with $s_t^c \leq q_t^c$, and (5) was enforced. The true next state is $z_{t+1} = f(z_t, a_t) + r_t$, so $\|z_{t+1} - (f(z_t, a_t) + U\hat{W}_c\phi_t)\| = s_t^c \leq q_t^c$. By the Lipschitz bound on h and (5), $h(z_{t+1}) \geq h(f + U\hat{W}_c\phi_t) - L_h q_t^c \geq (1 - \alpha)h(z_t)$. If no action satisfied (5), A5 gives the same conclusion. So the condition can fail only on steps that count as a miss for the attributed memory.

Misses per memory. For memory c with T_c attributed steps, the update gives $q_{\text{end}}^c - q_{\text{start}}^c = \gamma \sum (\mathbb{I}[\text{miss}] - \delta_s)$. The margin stays in $[-\gamma\delta_s, \bar{S} + \gamma]$: above $\bar{S}$ no miss is possible so the margin decreases, and below 0 every step is a miss so it increases. Hence the number of misses is at most $\delta_s T_c + (\bar{S} + \gamma + \gamma\delta_s)/\gamma \leq \delta_s T_c + (\bar{S} + 2\gamma)/\gamma$. Summing over the K_T slots and using $\sum_c T_c = T$ gives the result. $\square$

The argument uses no assumption on the data, which is why it holds with misleading cues. The bound applies to the safety condition in latent space. Whether this implies physical safety depends on whether h correctly represents the real constraint, which we treat as a property of the encoder [72]. In single-contact tasks it is natural to count episodes instead of steps, and the bound becomes a long-run rate of unsafe episodes. The term K_T grows with every created memory, which is one more reason to limit false creations (H7).

6.4 Unseen combinations

Proposition 4 (Unseen combinations). *[Proved here] Fix one output row and one input ϕ , and write $g(c) = w_c^\top \phi$. Under (2), $g(c) = \psi(c)^\top \beta + \xi_c$, where $\psi(c)$ stacks a constant 1 and one-hot indicators of $c^1, \dots, c^F$, β is Gaussian with positive definite diagonal covariance, and the $\xi_c \sim \mathcal{N}(0, \sigma_\times^2 \phi^\top \Sigma_0 \phi)$ are independent of each other and of β . Let Ω be the set of seen combinations, including the training condition, and suppose g is known exactly on Ω (the limit of many data per seen combination). Let $q \notin \Omega$ be a combination all of whose factor values occur in Ω .*

- (i) *If $\sigma_\times^2 = 0$, the posterior variance of $g(q)$ is zero if and only if $\psi(q) \in \text{span}\{\psi(\omega) : \omega \in \Omega\}$. For $F = 2$ this holds if and only if q^1 and q^2 lie in the same connected component of the bipartite graph that has an edge (ω^1, ω^2) for every $\omega \in \Omega$.*
- (ii) *If $\sigma_\times^2 > 0$, the posterior variance of $g(q)$ equals $\sigma_\times^2 \phi^\top \Sigma_0 \phi$ plus the posterior variance of $\psi(q)^\top \beta$ given the observations $g(\omega) = \psi(\omega)^\top \beta + \xi_\omega$, $\omega \in \Omega$. In particular it is at least $\sigma_\times^2 \phi^\top \Sigma_0 \phi$ however many combinations have been seen.*

Proof. (i) With $\sigma_\times = 0$ the observations are the linear functionals $\Psi\beta$, where Ψ has rows $\psi(\omega)^\top$. For a Gaussian β with positive definite covariance D , the posterior variance of $\psi(q)^\top \beta$ is the squared distance from $D^{1/2}\psi(q)$ to the range of $D^{1/2}\Psi^\top$, which is zero exactly when $\psi(q)$ lies in the row space of Ψ . For $F = 2$, suppose a path $q^1 = x_0, y_0, x_1, y_1, \dots, x_m, y_m = q^2$ alternates between factor-1 and factor-2 values along seen edges (x_j, y_j) and (x_{j+1}, y_j) . Then $\psi(q) = \sum_{j=0}^m \psi(x_j, y_j) - \sum_{j=0}^{m-1} \psi(x_{j+1}, y_j)$: the constant appears $m+1$ times with a plus sign and m times with a minus sign, and the interior indicators cancel. Conversely, if q^1 and q^2 lie in different components, let $\mathcal{C}$ be the component of q^1 and set the constant to 0, the factor-1 components to +1 on $\mathcal{C}$, the factor-2 components to -1 on $\mathcal{C}$, and all others to 0. Every seen edge lies entirely inside or entirely outside $\mathcal{C}$, so this β is orthogonal to every row of Ψ , while $\psi(q)^\top \beta = 1$. Hence $\psi(q)$ is not in the row space. (ii) ξ_q is independent of β and of all observations, so the posterior variance of $\psi(q)^\top \beta + \xi_q$ is the sum of the two parts. $\square$

The proposition has two uses. The conditional variance s_q at each unseen combination is computable from the fitted hyperparameters before the experiment, so we preregister the predicted first-contact error for H5. It also constrains the deployment schedule: the unseen-combination tests need a connected graph of seen combinations, and we include one deliberately disconnected pair as a control.

6.5 Calibrated memories under context uncertainty

Theorem 5 (Calibrated memories under context uncertainty). *[Proved here] Fix a memory c and an output coordinate i , and suppress i in the notation. Let*

$$w_t = \pi_{t|t-1}(c) / \sigma_i^2(z_t, a_t) \quad (\text{zero before the memory exists}), \quad V_t = P + \sum_{s < t} w_s \phi_s \phi_s^\top,$$

where P is the prior precision, and let the memory's posterior mean be

$$\hat{w}_t = V_t^{-1} \left(P m_t + \sum_{s < t} w_s \phi_s e_s \right),$$

where the prior centre m_t may be recomputed at any time. Let $\sigma_t(\phi) = \|\phi\|_{V_t^{-1}}$ be the posterior standard deviation of $\phi^\top w$. Under A1 and A2, with probability at least $1 - \delta$, simultaneously for all t , all $\phi \in \mathbb{R}^p$ and all k coordinates,

$$|\phi^\top (\hat{w}_t - w_c)| \leq \beta_t \sigma_t(\phi), \quad \beta_t = \sqrt{2 \log(k/\delta) + \log \frac{\det V_t}{\det P}} + \|m_t - w_c\|_P + B_t,$$

with the contamination term

$$B_t^2 = \sum_{s < t} w_s ((w_{c_s^*} - w_c)^\top \phi_s)^2.$$

If all inputs lie in $\mathcal{Z}_A$, then $B_t^2 \leq \Delta_{\max}^2 M_t$ with $M_t = \sum_{s < t: c_s^* \neq c} w_s$.

Proof. Set $x_s = \sqrt{w_s} \phi_s$, $\eta_s = \sqrt{w_s} \varepsilon_s$ and $d_s = \sqrt{w_s} (w_{c_s^*} - w_c)^\top \phi_s$. By A1 and A2, $\sqrt{w_s} e_s = x_s^\top w_c + d_s + \eta_s$. Substituting into $\hat{w}_t$ and using $V_t = P + \sum x_s x_s^\top$,

$$\hat{w}_t - w_c = V_t^{-1} \sum_{s < t} x_s \eta_s + V_t^{-1} \sum_{s < t} x_s d_s + V_t^{-1} P(m_t - w_c).$$

For any ϕ and u , $|\phi^\top V_t^{-1} u| \leq \|\phi\|_{V_t^{-1}} \|u\|_{V_t^{-1}}$ (Cauchy-Schwarz in the V_t^{-1} inner product). We bound $\|u\|_{V_t^{-1}}$ for the three terms.

Noise. This is where the choice of weights matters. The vector x_s is $\mathcal{F}_{s-1}$ -measurable, because ϕ_s and $\sigma_i(z_s, a_s)$ depend on (z_s, a_s) and $\pi_{s|s-1}$ depends only on data up to $s-1$ and the cue. Given $\mathcal{F}_{s-1}$, η_s is $\sqrt{\pi_{s|s-1}(c)}$ -sub-Gaussian, hence 1-sub-Gaussian. The self-normalised bound for vector-valued martingales [1, Thm. 1], applied with regulariser P and confidence δ/k , gives, simultaneously for all t , $\|\sum_{s < t} x_s \eta_s\|_{V_t^{-1}}^2 \leq 2 \log(k/\delta) + \log(\det V_t / \det P)$. With the filtered weight $\pi_s(c)$ in place of $\pi_{s|s-1}(c)$, x_s depends on ε_s , the sum is no longer a martingale transform, and this step fails.

Contamination. Let X have rows $x_s^\top$ and $d = (d_s)_s$. Then $\|X^\top d\|_{V_t^{-1}}^2 = d^\top X V_t^{-1} X^\top d \leq \|d\|^2 = B_t^2$, because $V_t \succeq X^\top X$ implies $X V_t^{-1} X^\top \preceq I$ (its nonzero eigenvalues are those of $V_t^{-1/2} X^\top X V_t^{-1/2} \preceq I$). This bound holds pathwise and uses nothing about how the weights were chosen.

Prior. $\|P(m_t - w_c)\|_{V_t^{-1}}^2 = (m_t - w_c)^\top P V_t^{-1} P(m_t - w_c) \leq \|m_t - w_c\|_P^2$, because $V_t \succeq P$ implies $P V_t^{-1} P \preceq P$. This also holds pathwise, which is why m_t may be recomputed from past data at any time.

A union bound over the k coordinates completes the proof. For the last claim, $d_s = 0$ when $c_s^* = c$ and $|(w_{c_s^*} - w_c)^\top \phi_s| \leq \Delta_{\max}$ otherwise. $\square$

Interpretation. The step usually skipped is the contamination bound: data from the wrong context enter with total weight M_t , and the inequality $X V_t^{-1} X^\top \preceq I$ says their effect on any prediction is at most $\Delta_{\max} \sqrt{M_t}$ posterior standard deviations. Since $\sigma_t(\phi)$ shrinks roughly like $1/\sqrt{n_t}$ in the effective sample size n_t , the contamination error behaves like $\Delta_{\max}$ times the square root of the fraction of the memory's (noise-weighted) data that came from other conditions. It

does not vanish with more data, and keeping it small is the job of the context filter. Summed over memories,

$$\sum_c M_{c,T} \leq \frac{1}{\sigma_{\min}^2} \sum_{t < T} (1 - \pi_{t|t-1}(c_t^*)) \leq \frac{1}{\sigma_{\min}^2} \sum_{t < T} -\log \pi_{t|t-1}(c_t^*),$$

the step-level log-loss of the context filter. The same kind of quantity controls the first-contact error in Proposition 1, at the episode level.

The prior term and negative transfer. [Heuristic] If the prior (2) is correct, $\|m_t - w_c\|_P$ is of order $\sqrt{p}$ whatever s_c is; the gain from sharing appears in the starting width $\sigma_0(\phi)^2 = s_c \phi^\top \Sigma_0 \phi$, which is small when the stored contexts explain the new one. If k_C overstates the sharing, $\|m_t - w_c\|_P$ is large and the bound widens. This is the negative-transfer failure mode. The coverage metric in Section 9 detects it, and per-memory ACI keeps the safety guarantee intact.

Drift. [Provable with effort] If the weights W_c drift, discounting old data restores a bound with an extra drift term, as in weighted linear bandits [88] and time-varying GP bandits [16].

Remark 1 (Filtered weights bias the memory). [Proved here] Take scalar errors, $\phi \equiv 1$, two memories with means 0 and Δ , true context 0, noise $\mathcal{N}(0, \sigma^2)$, equal prior probabilities, and an update that stores e_t in the memory with the larger filtered probability π_t . Memory 0 receives e_t exactly when $e_t < \Delta/2$, so the errors it stores have mean $\mathbb{E}[\varepsilon \mid \varepsilon < \Delta/2] = -\sigma \varphi(\Delta/2\sigma)/\Phi(\Delta/2\sigma) < 0$ for every Δ . Its estimate converges to this value, while its confidence interval shrinks like $\sigma/\sqrt{n}$, so the interval eventually excludes the true value 0. For $\Delta = 2\sigma$ the bias is about -0.29σ . Soft filtered weights reduce the bias without removing it, because the weight of memory 0 decreases in ε . With predictive weights the stored errors keep mean zero.

6.6 From prediction error to task cost

Lemma 6 (Simulation lemma). [Proved here] Let a plan be fixed, let z_t be the true trajectory and $\hat{z}_t$ the trajectory of a model $\hat{F}$, both from z_0 , and let $\hat{x}_t$ denote the model's state-action pair at step t . Under A3–A4,

$$|J - \hat{J}| \leq L_c C_H \sum_{t=0}^{H-1} \|U\mu_{c^*}(\hat{x}_t) - (\hat{F}(\hat{x}_t) - f(\hat{x}_t))\|.$$

Proof. Let $E_t = \|z_t - \hat{z}_t\|$ and let F^* be the true map. Adding and subtracting the true map at the model state, $E_{t+1} \leq \|F^*(z_t) - F^*(\hat{z}_t)\| + \|F^*(\hat{z}_t) - \hat{F}(\hat{z}_t)\| \leq LE_t + \text{err}_t$, where err_t is the summand above. Unrolling from $E_0 = 0$ gives $E_t \leq \sum_{s < t} L^{t-1-s} \text{err}_s$. Summing $L_c E_t$ over t and exchanging the sums gives the claim. $\square$

The choice in the proof is to evaluate model error along the model's own trajectory. That makes the bound computable at planning time.

Corollary 7 (First-contact decision gap). [Proved here, given A1–A4] In episode e , let the strike $\hat{a}$ minimise $\hat{J}(a) = \sum_c \pi_0^e(c) \hat{J}_c(a)$ over a candidate set $\mathcal{A}$, where $\hat{J}_c$ rolls out $f + U\hat{W}_c\phi$ for H steps after contact. Let $\bar{\varepsilon}_e$ bound $\|(\hat{W}_c - W_c)\phi\|$ over the contexts in the support of π_0^e and the inputs of their rollouts in $\mathcal{Z}_A$. Then

$$J(\hat{a}) - \min_{a \in \mathcal{A}} J(a) \leq 2L_c C_H H (\bar{\varepsilon}_e + (1 - \pi_0^e(c_e^*)) \Delta_{\max}),$$

and summed over episodes, $\sum_e (J(\hat{a}_e) - \min_a J(a)) \leq 2L_c C_H H (\sum_e \bar{\varepsilon}_e + \Delta_{\max} \sum_e -\log \pi_0^e(c_e^*))$.

Proof. For any a , $|\hat{J}(a) - J(a)| \leq \sum_c \pi_0^e(c) |\hat{J}_c(a) - J(a)|$. Apply Lemma 6 to model c : since U has orthonormal columns, its per-step error is at most $\|(\hat{W}_c - W_c)\phi\| + \|(W_c - W_{c^*})\phi\| \leq \bar{\varepsilon}_e + \mathbb{I}[c \neq c^*] \Delta_{\max}$. Hence $|\hat{J}_c(a) - J(a)| \leq L_c C_H H (\bar{\varepsilon}_e + (1 - \pi_0^e(c^*)) \Delta_{\max})$ for every a . With a^* the true minimiser, $J(\hat{a}) - J(a^*) \leq [J(\hat{a}) - \hat{J}(\hat{a})] + [\hat{J}(a^*) - J(a^*)]$, since $\hat{J}(\hat{a}) \leq \hat{J}(a^*)$. The sum over episodes uses $1 - x \leq -\log x$. $\square$

The second bracket lives on the trajectory of the unknown best strike, so the bound must hold uniformly over $\mathcal{A}$. It does, because Theorem 5 holds uniformly over inputs and $\Delta_{\max}$ is a supremum. Two caveats apply to the strike. The puck’s position integrates its velocity, so $L \geq 1$ and $C_H \geq H$, which makes the bound grow at least quadratically in the sliding horizon. The moment the puck stops is also not Lipschitz, which violates A3. In the state-based simulation we therefore check the corollary through the closed-form sliding distance $d = v_0^2/(2\mu_f g)$, whose first-order relative error is $|\delta d|/d \leq 2|\delta v_0|/v_0 + |\delta \mu_f|/\mu_f$.

6.7 When a probe has value

Proposition 8 (Probes are worthless when the plausible contexts agree on the action). *[Proved here] Let the state of nature θ (context and memory weights) have belief b , and let $J_\theta(a)$ be the expected cost of strike a under θ . For any probe p with outcome o , $\text{VOI}(p) \geq 0$. If some strike $a^\dagger$ minimises J_θ for every θ in the support of b , then $\text{VOI}(p) = 0$ for every probe, while $I(\theta; o | p) > 0$ whenever the outcome distribution differs between two such θ of positive probability.*

Proof. $V(b) = \min_a \mathbb{E}_b[J_\theta(a)]$ is a minimum of linear functions of b , hence concave, and $\mathbb{E}_o[b_o] = b$, so Jensen’s inequality gives $\mathbb{E}_o[V(b_o)] \leq V(b)$. For the second claim, every posterior b_o is supported inside the support of b , so $a^\dagger$ minimises $\mathbb{E}_{b_o}[J_\theta(\cdot)]$ for every o . Then $\mathbb{E}_o[V(b_o)] = \mathbb{E}_o \mathbb{E}_{b_o}[J_\theta(a^\dagger)] = \mathbb{E}_b[J_\theta(a^\dagger)] = V(b)$. The last claim is the definition of mutual information. $\square$

The proposition gives a testable difference between the two probing rules. An information-seeking rule probes whenever the plausible contexts predict different probe outcomes. The value-of-information rule probes only when the plausible contexts also call for different strikes. Its probing rate should therefore track disagreement between the best strikes of the plausible contexts and be unrelated to the entropy of π once that disagreement is controlled for (H9).

7 Measuring recall versus learning

For every adaptive method in this proposal, we separate its adaptive state into two parts: which memories it uses, π (for Cira this is the context distribution), and what the memories contain, Θ (the weights, including prior centres recomputed through (2)). Methods that reset every episode have only Θ . Let $L(\pi, \Theta)$ be the prediction error on data from the current condition, computed by replaying recorded transitions.

Definition 2 (Splitting recovery into recall and learning). After a change at time t_0 , the improvement by time t is $R_t = L(\pi_{t_0}, \Theta_{t_0}) - L(\pi_t, \Theta_t)$. The part due to changing which memories are used is

$$A_t = \frac{1}{2}(L(\pi_{t_0}, \Theta_{t_0}) - L(\pi_t, \Theta_{t_0})) + \frac{1}{2}(L(\pi_{t_0}, \Theta_t) - L(\pi_t, \Theta_t)),$$

the average effect of switching π with Θ held at its old and at its new value (a Shapley value with two players). We call A_t apparent learning. The remainder $P_t = R_t - A_t$ is proper learning.

The two parts always add up to the total, and the result does not depend on which part is computed first. For Cira every term can be computed in closed form from stored distributions. For methods that reset every episode, A_t is always zero. We apply the same split to the safety margin. This shows how much of the reduction in caution after a change comes from recalling a memory and how much from recalibration.

8 Hypotheses and stopping criteria

H1 First contact. On single-contact tasks with recurring conditions and an informative cue, Cira succeeds more often than the frozen model and all error-driven baselines. Its first-contact error follows Proposition 1, and its decision gap in the state-based simulation follows Corollary 7.

- H2 **Savings.** Recognition time falls with each further occurrence, as predicted by Proposition 2. For gradient updates with reset it does not change.
- H3 **No interference.** Learning condition B does not reduce performance on condition A. Continuously running gradient updates and continual test-time adaptation do reduce it.
- H4 **Misleading cues.** The observed dynamics overrule a misleading cue within the predicted number of steps. Task cost stays below that of the frozen model, and the safety bound holds.
- H5 **Sharing and new combinations.** For a combination of factors never seen together (for example the acrylic mat with the weighted puck), the shared prior gives lower first-contact error than both the version without cues and the version with independent memories. The observed error matches the preregistered s_q from Proposition 4, and it is larger for the deliberately disconnected pair. For new contexts that share no factor value with stored ones, the shared prior does no worse than independent memories.
- H6 **Safety.** The long-run rate of violations of the latent safety condition stays within δ_s plus the predicted slack, also with misleading cues. At the same violation rate, our filter intervenes less than a filter that guards against all memories and less than a filter with one shared margin. Its caution decreases as conditions recur.
- H7 **Noise.** With noisy contacts, no single learning rate for gradient updates is best both in normal conditions and after a change. Cira matches the best rate in each phase. Its rate of falsely created contexts stays below a limit fixed in advance.
- H8 **Recall versus learning.** In the long real deployment, recall accounts for a substantial share of recovery. Fixed in advance: the median of A/R is at least 0.3.
- H9 **Probing.** Value-of-information probing reaches the success rate of information-seeking probing with fewer probes. Its probing rate tracks disagreement between the best strikes of the plausible contexts; the rate of information-seeking probing tracks the entropy of π .
- H10 **Calibration.** With predictive weights, the empirical coverage of the intervals of Theorem 5 (with $\Delta_{\max}$ estimated from the stored memories) is at least $1 - \delta$ minus a tolerance fixed in advance. With filtered weights, coverage falls short, most strongly in phases where contexts overlap (small Δ/σ), with a bias of the size Remark 1 predicts.

When we change the plan.

- If the version without cues is as good as the full method at first contact, we drop the claim about acting before contact. The paper then focuses on memory, the benchmark and the recall/learning measurement.
- If well-tuned gradient updates with reset are as good as the version without cues on recurring conditions, the method offers no advantage. The paper then reports the benchmark and this negative result.
- If the continuous-context model (Section 9) matches Cira at first contact, discrete memories are not needed. We then report this and restrict the claims to the context filter, the shared prior and the safety filter.
- If the shared prior fails on unseen combinations, we report independent whole-context memories only and drop the claim about new combinations.
- If value-of-information probing is not better than information-seeking probing at equal success, probing returns to an optional extension.
- If coverage stays below the preregistered level even with predictive weights, the noise model (A2) is the first suspect. We report coverage per phase and rely on the ACI guarantee alone.
- If our safety filter is not less cautious than the all-memories filter at the same violation rate, we present the safety part as a supporting result.

9 Experiments

Sequences of conditions. Each condition lasts for a random number of episodes (geometric distribution), and it occasionally changes within an episode. Each condition has a matching appearance. The cue shows that appearance with probability ρ and a different one otherwise. We test $\rho \in \{1/M, 0.5, 0.65, 0.8, 0.95\}$, where M is the number of conditions and $\rho = 1/M$ means the cue carries no information. The sequences contain four patterns taken from motor-learning experiments: savings (conditions A, B, A, B in blocks); interference (a long or a short block of A before B); evoked recovery (two episodes of A inserted into a block of normal conditions); and misleading cues (the pairing between cue and condition swapped for one block). Each pattern is linked to one of the predictions above. A pattern that does not separate the methods in simulation is removed. Schedules are designed so that the graph of seen combinations is connected before each unseen-combination test, except for one control pair (Proposition 4).

Environments.

- (i) *Simulation.* A 2D pushing task and a 2D puck strike with a table edge. Contacts are noisy, and surface and load vary. Cues are correct 80% of the time. These tasks run on a laptop and decide whether H1, H2, H4, H5, H6, H9 and H10 hold before we invest in hardware. The strike has a known state, so Corollary 7 can be checked quantitatively here.
- (ii) *AdaJEPA tasks* [102]. PushT, PushObj and PointMaze with the released JEPA and DINO-WM [110] models. We pair the existing colour changes of the agent or block with changes in dynamics, so that colour becomes a cue, and we add random noise to the dynamics. This allows a direct comparison with AdaJEPA and Sandwich-Residuals.
- (iii) *Real robot, single-contact tasks.* Strike: strike a puck to a target on felt, rubber or acrylic mats, without it sliding off the table edge [108]. Lift and place: lift loads in visually distinct containers without exceeding a limit on jerk or tilt. The world model is a frozen V-JEPA 2-AC [8] or a DINO-WM-style model trained on training-condition data. The memories add only a few kilobytes on top of a large pretrained model that stays unchanged.
- (iv) *Real robot, long deployment.* At least 1000 episodes over several days. People swap mats and containers according to a schedule the robot does not know, and sometimes cover the mats with a neutral sheet to make cues less reliable. We report the recall/learning split, predicted versus observed first-contact error, coverage, and the violation rate.

Baselines.

- *Adaptation:* the frozen model; continuously running gradient updates (as in AdaJEPA); gradient updates with periodic reset [85]; gradient updates reset exactly at each change (an idealised version); Sandwich-Residuals [94]; a MOLe-style mixture of models [79]; MOCA-style change detection without recall [41].
- *Continuous-context model* (from Direction 1): a Gaussian process on the residual with a product kernel $k_Z \otimes k_E$, where the environment descriptor e combines the cue embedding with a short context window of recent transitions, and no discrete memories. This tests whether discrete memories are needed.
- *Safety:* the uncertainty-aware latent filter [89]; a latent filter with one shared ACI margin [52]; a filter that guards against all memories [71].
- *Probing:* no probing; information-seeking probing $\arg \max_p I(c; o | p)$ (v3); value-of-information probing; the linearised c -optimal rule alone.
- *Ablations of Cira:* without cues; with independent memories (no shared prior); with filtered update weights (v3); with constant noise; with a constant-offset memory; updating only the most likely memory; with one shared margin instead of one per memory.

- *Upper bound:* Cira told the true context.

All settings are tuned on training-condition data only. Random seeds and sequences are fixed and published before the main experiments.

Metrics.

- Error at first contact, and success on single-contact tasks; predicted versus observed values for Propositions 1 and 2 and, in the state-based simulation, Corollary 7.
- Number of steps after contact needed to recognise the condition; savings as the recognition time for the k -th occurrence relative to the first.
- Total extra prediction error after each change; performance on condition A after learning condition B.
- Error on unseen combinations of factors against the preregistered s_q ; s_c for every new context.
- Empirical coverage of the intervals of Theorem 5; the contaminated weight fraction M_c/n_c per memory.
- Violation rate, and how much the safety filter changes the planned action or raises task cost.
- Probing rate against strike disagreement and against the entropy of π ; success per probe.
- The recall share A/R , for prediction error and for the safety margin.
- Rates of falsely created and missed contexts.

10 Risks

- **Contact noise creates false contexts.** This is the most likely problem. The state-dependent noise model and a minimum number of steps before a new memory is created address it. We report the rate of false contexts as a main metric.
- **Cues are weak.** We vary cue reliability. Proposition 1 predicts how much a weak cue should still help, so weak cues still yield a measurable result.
- **Factors interact.** Mass and friction are expected to interact. The term σ_x^2 in (2) absorbs the interaction, and Proposition 4 says how much error remains at unseen combinations.
- **Few contexts for the shared prior.** With a handful of contexts the hyperparameters of (2) are poorly determined, and overstated sharing causes negative transfer at the first contact of a new condition. We use weakly informative priors, report the posterior spread, and rely on the coverage metric to detect overconfidence. Per-memory ACI keeps the safety bound intact in this case.
- **The noise model underestimates noise.** Theorem 5 then fails. Coverage per phase detects this, and the safety guarantee does not depend on it.
- **Predictive weights delay learning.** The step at which a change occurs is stored with the old memory. We measure the effect on recognition time and on M_c .
- **Cost of probing decisions.** Eq. (4) is restricted to the elite strikes and to context reweighting, so it reuses rollouts the planner already computed.
- **A change falls outside the correction directions.** The safety margin is measured in the full latent space, so safety does not depend on the chosen directions. For prediction, we monitor the error outside these directions and add directions for a new memory when this error grows.

- **Latent safety differs from physical safety.** Proposition 3 concerns the latent condition. We report physical violations separately. Whether h matches the real constraint is outside the scope of this proposal [72].
- **Competing work.** AdaJEPA and Sandwich-Residuals name persistence as future work, and the latent ACI filters [52, 82] appeared this year. Our distinguishing points are acting before contact, the shared factor prior, calibration under context uncertainty, decision-aware probing, and margins stored per memory. A workshop paper by February 2027 establishes priority.
- **The motor-learning patterns add nothing.** Each pattern is tied to a numerical prediction and is removed if it does not separate the methods.

11 Timeline and division of labour

When	Work	Done when
Oct 2026	Simulated pushing and puck strike; core of Cira with shared prior and predictive weights; gradient baselines; coverage check for Theorem 5	H1, H2, H4, H10 visible; stopping criteria checked
Nov 2026	Unseen combinations with preregistered s_q ; safety filter with per-memory margins; value-of-information probing; recall/learning measurement; continuous-context baseline	H5, H6, H9 in simulation; Propositions 1–8 compared with simulation
Dec 2026–Jan 2027	AdaJEPA tasks with all baselines; real-robot setup (mats, containers, puck, edge)	Condition sequences running in PushT and PointMaze
Feb 2027	Workshop paper: simulation, AdaJEPA tasks, theory	Submitted
Mar–Apr 2027	Real-robot single-contact tasks; long deployment (≥ 1000 episodes); probing on hardware	H1–H10 tested on hardware
Spring 2027	Full paper for CoRL 2027 (deadline to be confirmed when announced)	Submitted

With three people: one on the theory (Propositions 1–8, the preregistered predictions, and the coverage and decision-gap checks); one on the simulated and AdaJEPA experiments with all baselines, including the continuous-context model; one on the real-robot infrastructure, the deployment schedule, and the prior-art checks. Value-of-information probing is the natural joint component with the Stanford side.

12 Prior art to settle before committing

- **Transductive selection and multi-task transfer in the LAS group** [9, 51]. Check whether either uses a task kernel over discovered contexts or a decision-defined target. Both are next to us, so a direct conversation is the cheapest check.
- **How COIN weights memory updates** [44]. If it uses filtered probabilities, Remark 1 applies to it, and the text must state our deviation explicitly.
- **Weighted and misspecified self-normalised bounds** [1, 61, 88]. Theorem 5 is a direct combination of known tools; the contribution is its use with weights set by a context filter, and the text should claim no more.
- **Value of information for probing in multiple-model control** [13, 14, 46, 91]. Active fault diagnosis already discriminates between models; check whether any of this work uses the decision, as opposed to model identity, as the target.
- **Identification for control** [17, 36, 47]: prior art for the linearised c -optimal probe.

Optional: safe probing beyond single contacts

In multi-contact tasks the same rule applies at every step: among candidate actions that satisfy (5) and whose task cost is within a factor $(1 + \epsilon)$ of the best, CEM favours those with the highest value of information. This connects to active fault diagnosis [13, 91], dual control [46, 78] and safe exploration [12, 58]. How often the robot chooses to probe is itself a result we report.

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